LUONG TIEN DUNG

AI / MACHINE LEARNING / DATA SCIENCE INTERN

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GitHub: github.com/Luong-dung | LeetCode: leetcode.com/u/__covetot__

Third-year Data Science student focused on PyTorch video classification, data preparation, and reliable validation, with complementary experience in C++ algorithm design and benchmarking.

EDUCATION

VNU University of Science (HUS), Vietnam National University, Hanoi
Bachelor's in Data Science | Expected 2028 | GPA: **3.11/4.00**

TECHNICAL SKILLS

Programming: Python, SQL, C++
ML / Vision: PyTorch, PyTorch Lightning, OpenCV; computer vision, machine learning
Data: Pandas, NumPy; preprocessing, augmentation, model evaluation
Algorithms / Tools: Data structures, search, optimization; Git, GitHub, Linux, Jupyter Notebook

SELECTED PROJECTS

Sign Language Recognition (Video Classification)

Independent experiments | Python, PyTorch, OpenCV

- Built an **end-to-end PyTorch pipeline** for a **100-class video classification** task, from frame sampling and preprocessing to augmentation, training, and inference.
- Experimented with **pretrained 2D CNN backbones** for visual features and **Transformer-based temporal modeling** across video frames.
- Investigated **duplicate samples, validation leakage, and class imbalance** to assess data quality and the reliability of validation results.
- Achieved **74.57% Macro-F1 (0.7457) on the private test set** of the **2025 OLP AI task** used for practice; also evaluated accuracy and macro F1-score.

Large-Scale N-Puzzle Solver

University group project | C++ | Algorithm design & benchmarking

- Designed a **macro-based strategy** with alternating row/column decomposition and BFS fallback for exceptional configurations.
- Benchmarked an **AI-optimized solver** on **1000 × 1000** boards: **21/30 solved (70%)** within **10 seconds / 1 GB RAM**; successful runs averaged **9.00 s**.
- Compared BFS, IDA*, and hybrid search; directed **AI-assisted implementation** and iterative optimization of the strategy.

COMPETITIVE PROGRAMMING & AWARDS

Third Prize (2025) — Vietnam Student Informatics Olympiad, Specialized Division
Honorable Mention (2024) — Vietnam Student Informatics Olympiad, Specialized Division
LeetCode: Contest Rating **1807** · **280** problems solved | **AtCoder:** Peak Rating **1610**

Available part-time during the academic year; open to full-time work in summer 2027.